

Soham Chitnis

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Education

- New York University**, Courant Institute of Mathematical Sciences
Master of Science in Computer Science (GPA: 3.95/4.0)
• Courses: Deep Learning, Natural Language Processing (NLP), Operating Systems, Computer Vision, DevOps
- New York, NY, USA
September 2024 - May 2026
- Birla Institute of Technology and Science Pilani (BITS Pilani)**
Bachelor of Engineering in Computer Science, Minor in Data Science
• Courses: Machine Learning, Reinforcement Learning, Database Systems, Data Structures & Algorithms
- Goa, India
November 2020 – July 2024

Experience

- CILVR Lab, New York University**
Research Assistant
• **Advisors:** Wancong (Kevin) Zhang, Prof. Yann LeCun
• Training Latent space World Models: JEPA and Hierarchical JEPA with self-supervised learning on robot (state-action) trajectories for latent space dynamics model, with Model Predictive Control (MPC) for planning.
- New York, NY, USA
January 2025 – Present
- APP Center for AI Research, BITS Pilani**
Research Assistant
• **Advisors:** Prof. Ashwin Srinivasan, Prof. Tanmay Verlekar
• Developed a **state-of-the-art** deep learning model for brain tumor classification on histopathology images, resulting in 96.85% AUC ROC, improving diagnostic confidence to 93.57% for reliability, and generated interpretive heatmaps for pathologists, earning an **oral presentation** and publication at IEEE EMBC 2023. Collaborated with Doctors from Macquarie University.
• Designed AutoRef, a **LLM multi-agent system** for reviewing research papers using LangChain, OpenAI text embedding models & prompt engineering, yielding improvement in human reviews by 217% and AI-generated reviews by 15%. Collaborated with TCS Research. Published & presented at **ACL 2024 @ SDProc Workshop**.
- Goa, GA, India
February 2022 – May 2024
- Tata Consultancy Services Research**
Research Intern
• **Advisor:** Dr. Manasi Patwardhan
• Engineered an efficient multimodal model for chart understanding by grounding a frozen LLM with a trainable linear adapter, then analyzed the impact of pretraining methods for vision encoders (CLIP, DePlot) with language supervision on ChartQA.
- Pune, MH, India
September 2023 – December 2023
- VC Lab, Ontario Tech University**
MITACS Globalink Research Intern
• **Advisor:** Prof. Faisal Qureshi
• Engineered a novel spatial attention CNN encoder Latent Dirichlet VAE for Hyperspectral Unmixing, resulting in more than 95% reduction in standard deviation of prediction, with this work earning a publication & an **oral presentation** at IEEE IGARSS 2024. Awarded the **MITACS Globalink Research Internship Scholarship** to pursue research in Canada.
- Oshawa, ON, Canada
June 2023 – August 2023

Skills

Languages: Python, C++ , C, Java, JavaScript, Bash, SQL **Web & API Development:** REST APIs, FastAPI, HTML/CSS
Machine Learning: PyTorch, TensorFlow, JAX, Scikit-Learn, Pandas, NumPy, Hugging Face, LangChain, DSPy, OpenCV, MLFlow
Developer Tools & DevOps: GIT, Docker, CI/CD, AWS, Linux, Slurm, Prefect
Software & Scientific Computing: MATLAB, LaTeX, MS Office Suite (Excel, Word, PowerPoint)

Selected Projects

- Self-supervised learning with Latent Space World Model (JEPA)**
• Explored a diverse set of self-supervised learning methods (two-step & joint training) for training latent space world models (JEPA), resulting in a state-of-the-art probing loss of 0.84, culminating in a presentation as a top-8 selected team.
- Presentation
- Action Conditioned Representation Alignment (AC-REPA) for Navigation World Models**
• Computer Vision Course Project advised by Prof. Saining Xie
• Working on improving temporal dynamics (physics understanding) in Conditional Diffusion Transformer (CDiT) using Representation Alignment (REPA) with Video Foundational Models trained using Self-supervised learning.
- Real-time Credit Card Fraud Detection**
• Architected and implemented an end-to-end MLOps pipeline for a real-time credit card fraud detection system, utilizing MLFlow tracking server on AWS EC2 and a PostgreSQL database on AWS RDS for remote experiment tracking and model registry.
• Performed Exploratory Data Analysis for feature selection and trained models with a workflow orchestration tool, Prefect, achieving F1-score 73.5%, and monitored Data Drift and metrics with EvidentlyAI.
• Deployed a web server for model serving using AWS ECR, Docker, and FastAPI on EC2, with a CI/CD pipeline in GitHub Actions that automated a comprehensive suite of unit and integration tests, pushing validated container images to ECR.
- Code

- Fine-tuned Multi-modal Foundational Model (ChartMoE) using LoRA, BFloat16 quantization, Flash Attention2 on SciVQA Dataset. Achieved Avg F1 score (Rouge-1, Rouge-L & BertScore): 0.7970.

MLOps Zoomcamp Assignments

[Code](#)

- Trained ML models for NYC Taxi ride duration prediction with an end-to-end MLOps Pipeline in batch processing, streaming with AWS Kinesis, and a web service with Flask. Deployment using Docker and Terraform for Infrastructure as Code.

Video Vision Transformers (ViViT) for 3D Medical Images

- Implemented Video Vision Transformer (Original Implementation in JAX) for 3D Medical Images: CT scans in PyTorch.

Operating System Course assignments

[Code](#)

- Implemented fundamental OS modules in C++, including a linker to combine object modules, a virtual memory manager (MMU) with paging, different page replacement algorithms, and a diverse set of process schedulers and I/O schedulers.

Publications

- AutoRef: Generating Refinements of Reviews Given Guidelines
Chitnis S., Patwardhan, M., Srinivasan, A., Verlekar, T.T., Vig, L., Shroff, G. SDP Workshop, ACL 2024
- SpACNN-LDVAE: Spatial Attention Convolutional Latent Dirichlet Variational Autoencoder for Hyperspectral Pixel Unmixing
Chitnis S., Mantripragada, K., Qureshi, F.Z. **(Oral)** IEEE IGARSS 2024
- Domain-Specific Pre-training Improves Confidence in Whole Slide Image Classification
Chitnis S.R., Liu, S., Dash, T., Verlekar, T.T., Di Ieva, A., Berkovsky, S., Vig, L., Srinivasan, A. **(Oral)** IEEE EMBC 2023

Volunteer

Ethics Reviewer NeurIPS 2025, ACL 2025 (Scholarly Document Processing Workshop), NeurIPS 2025 Position Track Reviewer

Honors & Awards

MITACS Globalink Research Internship at Ontario Tech University2023

100 percentile in Maharashtra (State-level) Engineering Entrance Test2020

(Physics-Chemistry-Mathematics & Mathematics)